

FR-017 M1 — Orchestration Observability Contracts

FR017 — sources/optional/m1.md

Masterclass source package snapshot

Functional Requirement	FR-017
Edition	Engineering Learning Academy — Source Package snapshot
Status	**Complete (M1)** — contracts frozen under accepted narrow scope
Audience	Academy generation and study (snapshot; SoT remains authoritative)
Document type	Source package snapshot (PDF)

Faithful rendering of a regenerable Masterclass source snapshot. Do not hand-edit; regenerate via python scripts/build_masterclass_package.py. Repository documentation remains the source of truth.

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- 3.4 API surface (M1)

FR-017 M1 — Orchestration Observability Contracts

Status: Complete (M1) — contracts frozen under accepted narrow scope (**Accepted** — unlocked M2); FR-017 later **Complete / Frozen** — [fr017_agent_evaluation_observability.md](#)

Date: 2026-08-07

Phase: Horizon 1A Stage 11

Architecture: [ADR-009](#) (Accepted)

Preceding: [M0 engineering spike](#) (**Accepted** — owner GO under narrow derive-only scope)

Succeeded by: [M2 corpus reconstructability](#) (**Complete — GO**)

Does not begin: M3 CLI, dashboards, DOS / BOPA / OBS runtime changes, Horizon 1B.

1. Executive summary

M1 delivers a **derive-only** observability contract layer in

`career_intelligence.multi_agent.observability`:

- deterministic per-run and corpus metrics over existing `OrchestrationRun` + `Handoff` audits
- optional child `AgentRunMetrics` roll-up via frozen FR-015 helpers (passed in; no `AgentRun` store reads inside the module)
- R1–R12 reconstructability evaluation helpers
- parent ↔ handoff ↔ child correlation with explicit orphan detection
- normative **missing vs zero** semantics for optional provider / token / cost / latency fields

No dashboards, event streams, telemetry stores, tracing platforms, CLI presentation, or FR-017 runtime changes. **Horizon 1B remains unblocked.**

Contract blocker check: No genuine gap against R1–R12 required changing DOS, BOPA, OBS, Handoff, `AgentRun`, or orchestration runtime. Derive-only gate **passes**.

2. Architectural decisions

Decision	Choice	Why
Package	<code>multi_agent.observability</code>	Mirror FR-015 <code>agent.observability</code> ; stay beside FR-016 audits
Derive-only	Pure functions; handoffs passed in	No new SoT; no store coupling in M1
Child metrics	Reuse <code>AgentRunMetrics</code> / caller supplies	Do not fork FR-015
Reconstructability	First-class <code>evaluate_reconstructability</code>	M0 R1–R12 become executable contracts
Runtime mutation	Forbidden in M1	Owner constraint; no blocker proven
CLI / corpus suite	Deferred to M2/M3	Contracts-only milestone
Horizon 1B	Not gated	M0 §9 normative

```
OrchestrationRun + Handoffs (+ optional AgentRunMetrics)
  |
  ▼
multi_agent.observability ← M1 (pure, read-only)
  ├── OrchestrationRunMetrics / HandoffMetrics
  ├── ParentChildCorrelation
  ├── OrchestrationCorpusMetrics
  └── ReconstructabilityReport (R1–R12)
```

3. Deterministic metrics contracts

3.1 Per-run — `OrchestrationRunMetrics`

Derived from one `OrchestrationRun` plus the handoff records the caller loads (typically via existing `handoff_ids`). Notable fields:

Family	Fields (representative)
Identity / goal	<code>orchestration_run_id</code> , <code>opportunity_id</code> , <code>owner_goal_label</code> , <code>goal_kind</code> , <code>flags</code>
Outcome	<code>status</code> , <code>stop_reason</code> , <code>owner_action_required</code>
Limits	<code>step_count</code> / <code>max_steps</code> , visit records, <code>step_limit_reached</code> , <code>visit_limit_reached</code>
Handoffs	counts; allow/deny; per-handoff <code>HandoffMetrics</code>
Observation snapshot	last hash, package/truth/pipeline/readiness class
Children	<code>child_agent_run_ids</code> , optional <code>child_agent_metrics</code> , <code>ParentChildCorrelation</code>
Optional provider roll-up	<code>provider</code> , <code>model</code> , <code>tokens</code> , <code>cost</code> , <code>elapsed_ms</code>

3.2 Per-handoff — `HandoffMetrics`

Selection reason, policy allow/deny (+ deny reason), acceptance lifecycle, observed hash, idempotency key, expected output, child refs, optional `handoff_elapsed_ms` (None when unresolved).

3.3 Corpus — `OrchestrationCorpusMetrics`

Aggregates over many `OrchestrationRunMetrics`: run/step/handoff totals, stop reason histogram, specialist visit totals, optional token/cost/elapsed totals.

3.4 API surface (M1)

Function	Role
<code>extract_handoff_metrics</code>	One handoff → <code>HandoffMetrics</code>
<code>correlate_parent_child</code>	Parent/child gap detection
<code>extract_orchestration_run_metrics</code>	Run + handoffs (+ children) → metrics
<code>aggregate_orchestration_metrics</code>	Corpus roll-up
<code>evaluate_reconstructability</code>	R1–R12 report

4. Missing versus zero (normative)

Field class	Empty audit	Measured zero
Counts (<code>handoff_count</code> , <code>allow/deny</code> , <code>events</code> , <code>steps</code> , ...)	<code>0</code>	<code>0</code>
Optional provider / model / tokens / cost / latency	<code>None</code> (absent)	<code>0 / 0.0</code> means measured zero

Never coerce `None` → `0` for optional provider metadata. Offline / no-child runs keep token and cost fields `None`. Corpus aggregates keep total token/cost/ elapsed as `None` when no constituent supplied a value.

Unit coverage: `tests/unit/multi_agent/test_observability_m1.py`.

5. Parent / child correlation semantics

`ParentChildCorrelation` compares:

- parent `child_agent_run_ids` ↔ handoff `child_agent_run_ids`
- parent `last_brief_id` ↔ handoff `child_brief_id`s
- handoff `orchestration_run_id` must match parent

Orphans are listed explicitly (`orphan_parent_*` , `orphan_handoff_*` , `orphan_handoff_brief_ids`).

`correlation_complete` is true only when there are no orphans and all handoffs cite the parent run id.

R11 consumes this structure.

6. R1–R12 reconstructability contracts

Implemented as `evaluate_reconstructability` → `ReconstructabilityReport`.

Id	Question (summary)	Evidence used
R1	Owner goal	OrchestrationGoal
R2	Authoritative observation	last_observation (+ hash)
R3	Specialist selection / why	allow handoffs, deny handoffs, or selection events
R4	DelegationPolicy allow/deny	handoff policy_* consistency
R5	Specialist authority boundary	target $\in$ { obs , bopa }
R6	Handoff lifecycle	acceptance present
R7	Child AgentRun / brief	completed allow handoffs cite expected child refs (stopped may omit)
R8	Stop reason	required when not running
R9	Owner next step	required when awaiting_owner
R10	Step / visit limits	limits visible; flags report proximity/hit
R11	Parent → handoff → child walk	ParentChildCorrelation (vacuous OK if no children/handoffs)
R12	Resume / idempotency evidence	observation hashes and/or idempotency keys (early running with no handoffs vacuous)

Documented early-run limitation: A brand-new `running` run with no handoffs and no selection events may fail R3 until selection evidence exists. That is honest reconstructability, not a runtime defect. Corpus demonstration of R1–R12 across FR-016 cases A–T is **M2**.

Optional `prior_observation_hash` strengthens R12 when comparing resume observations.

7. Ownership boundaries

Concern	Owner	FR-017 M1 role
Workflow traces / checkpoints / retries	FR-008	None — do not re-instrument
Unsupported-claim / truth engine	FR-014	None — observation may <i>cite</i> truth status only
BOPA metrics / AgentRun corpus	FR-015 (<code>agent.observability</code>)	Reuse — never redefine
Parent/child audit, handoffs, limits	FR-016	Source audits only
Orchestration metrics + R1–R12 helpers	FR-017	Derive / aggregate / evaluate only
Dashboards / SaaS tracing	—	Out of scope
Horizon 1B recruiter work	FR-018+	Not blocked on FR-017

8. Explicit out-of-scope (M1 and remaining FR-017)

- Dashboards, chart UIs, SaaS observability backends
- New event streams, telemetry stores, tracing platforms
- New orchestration audit event kinds
- DOS / BOPA / OBS / Handoff / AgentRun / DelegationPolicy / ToolPolicy changes
- FR-008 re-instrumentation; FR-014 claim duplication
- CLI metrics presentation (candidate M3)
- Corpus reconstructability suite wiring (M2)
- Browser / Playwright journey evidence
- Making `cic agent orchestrate` the daily prep default
- Blocking Horizon 1B on FR-017 completion

9. Derive-only gate (M1 result)

Criterion	Result
Every M1 metric is a pure function of stored audits (+ optional child metrics)	Pass
Zero new runtime event kinds	Pass
Zero DOS/BOPA/OBS/Handoff/AgentRun behaviour changes	Pass
Contract blocker requiring runtime change	None proven

10. Definition of Done (M1)

Criterion	Status
Deterministic metrics contracts	Done (<code>observability.py</code>)
R1–R12 helpers	Done
Parent/child correlation semantics	Done
Missing vs zero semantics	Done + tested
Optional provider/token/cost fields	Done (roll-up; None when absent)
Ownership boundaries documented	Done (§7)
Out-of-scope explicit	Done (§8)
Unit tests	Done (<code>test_observability_m1.py</code>)
ADR-009	Done
No dashboards / CLI / runtime changes	Confirmed
Stop before M2	Confirmed

11. Owner next step

Done: Owner accepted M1; M2 complete ([fr017_m2_corpus_reconstructability.md](#)).

Next: Accept M2 GO to unlock optional **M3** (thin metrics CLI) — or defer remaining FR-017 and proceed to Horizon 1B (allowed).

Do not implement M3 without explicit owner request.